Deep Dive into Git

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By the end of the session:

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You will get a hands-on understanding of managing your project with Git and implement most common Git commands.

Today's Learning Objectives

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- Understanding of managing your project with Git
- Implementing most common Git commands

Poll Time

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Q. Which of the following is not a component of Git?

a. Working Directory

b. Staging Area

c. Data Storage

d. Remote Repository



Poll Time

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a. Working Directory

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d. Remote Repository



Basic Git Commands

Git Init

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`git init`

This command initializes an empty git repo.



Git Status

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```
git status
```

This command gets the current status of the repo.



Git Add

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`git add`

This command adds the files to the staging area.

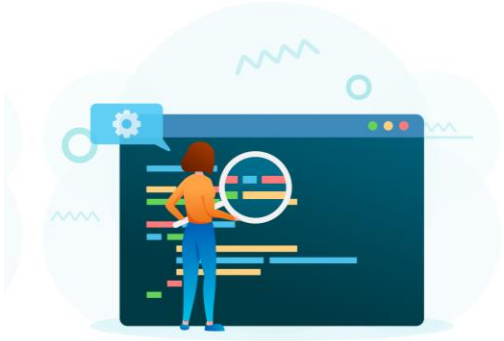


Git Commit

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`git commit`

This command commits the changes.



Git Diff

`git diff` is the universal diff command used for comparison.

This command helps in listing all the changes between your current working directory and the staging area.

```
$ git diff <commit1> <commit2>
```



Rolling Back

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`git revert`

It reverses the changes made in the latest **commit** option. The lines added will be removed, the removed lines will be added, and at the end, **commit** option will be used again.



Poll Time

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Q. Does git revert create a new commit?

a. Yes, it creates a new commit which rolls back changes of the last commit

b. No, it just rolls back the changes without polluting the commit history



Poll Time

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Q. Does git revert create a new commit?

a. **Yes, it creates a new commit which rolls back changes of the last commit**

b. No, it just rolls back the changes without polluting the commit history

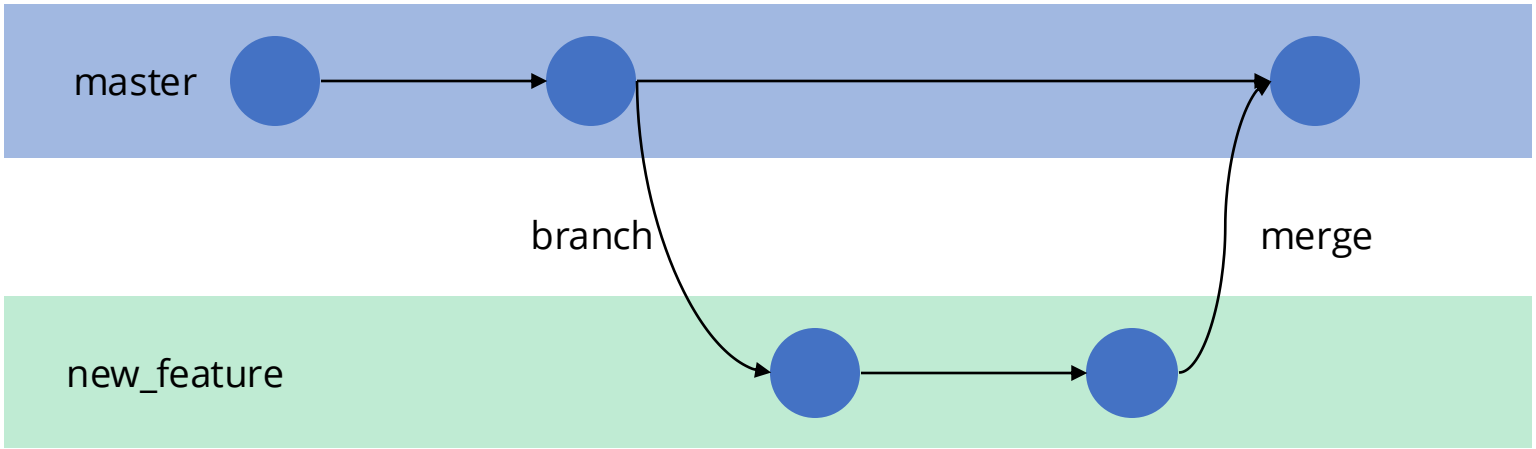


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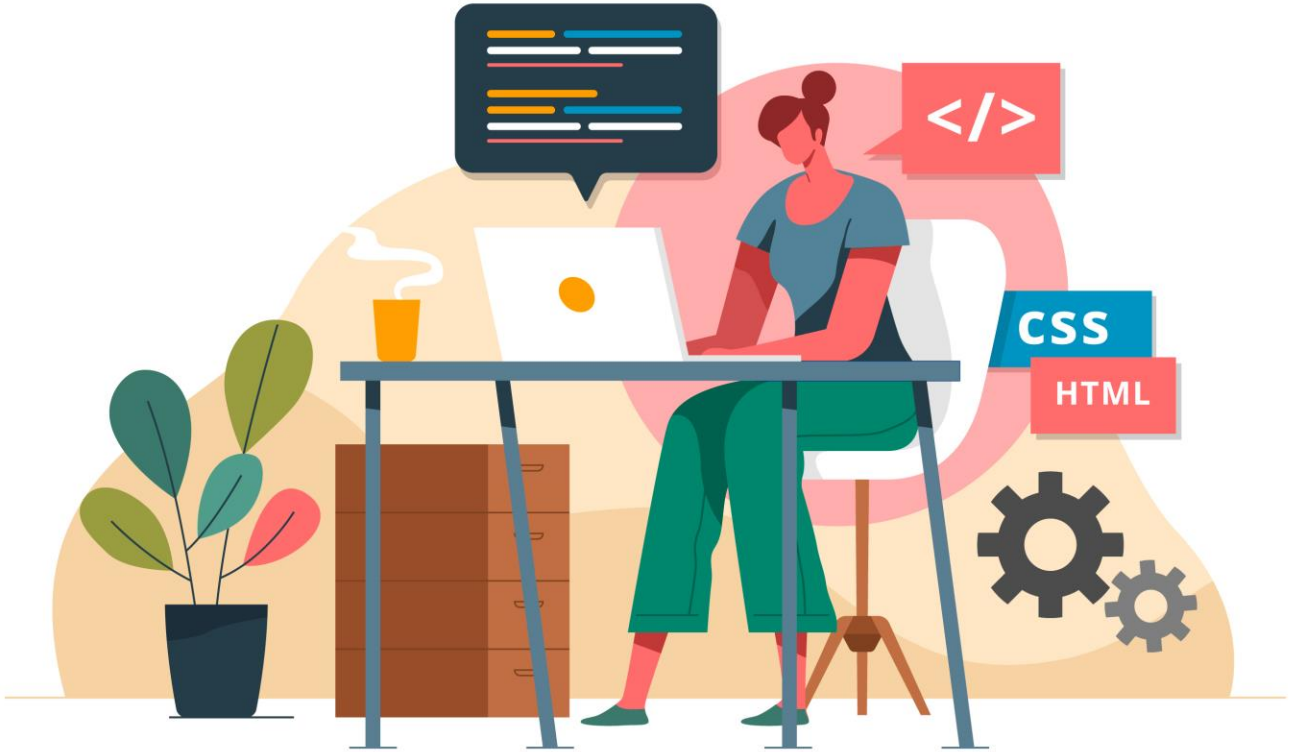
Branches

Git Branches



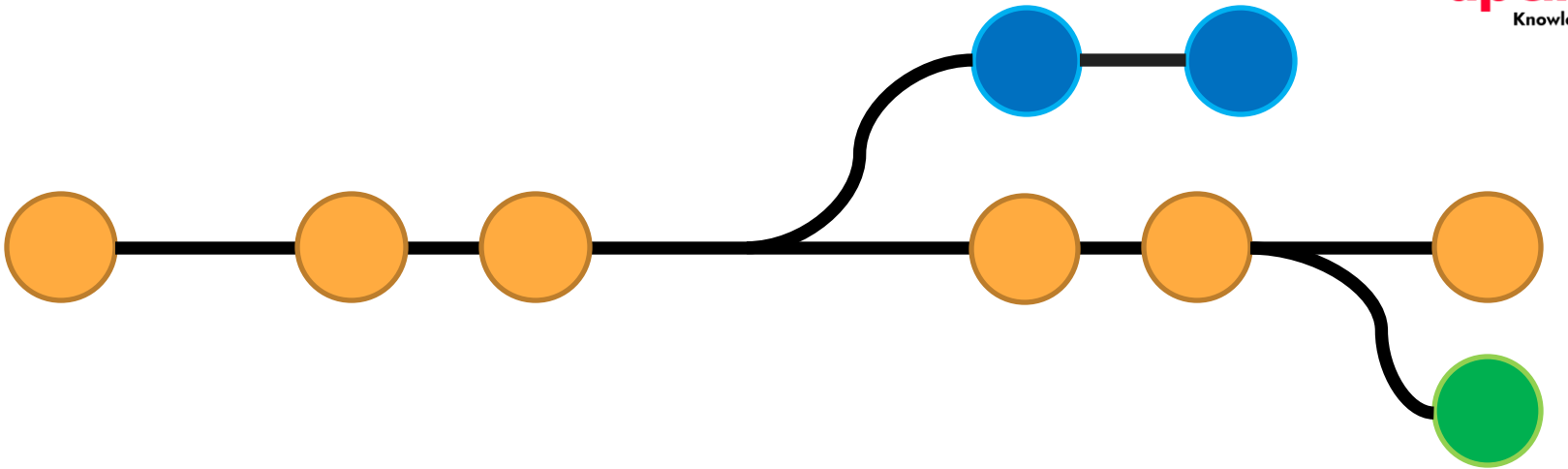
Challenges Faced by Developers before Branches

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Branches

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When you need to develop a new feature, fix an issue, or create an enhancement, create a branch and keep working on that branch without affecting the main version control track (branch).

When your fix or feature is ready, the updates from the branch can be merged into the main branch, thus releasing your feature for primary usage. This way, branches help you develop easily without affecting or damaging your application's primary line of development.

Creating and Checking Out a Branch

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To create a new branch, use the git branch command. The command below will instantiate a new branch named **notifications_feature:**

```
$ git branch notifications_feature
```

To start working with this new branch, use **git checkout** as shown below:

```
$ git checkout notifications_feature
```

Or you can save the effort and achieve the above in one step:

```
$ git checkout -b notifications_feature
```

This will create a new branch and checkout as well in one step!

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Pop Quiz

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Q. Which Git command is used to stage changes for a commit?

a. git add

b. git commit

c. git push

d. git pull



Pop Quiz

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Q. Which Git command is used to stage changes for a commit?

a. **git add**

b. git commit

c. git push

d. git pull



Important Commands

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To list all branches, use:

```
$ git branch
```

To delete a branch, use the following notation. This will not work if you have unmerged updates in the branch.

```
$ git branch -d notifications_feature
```

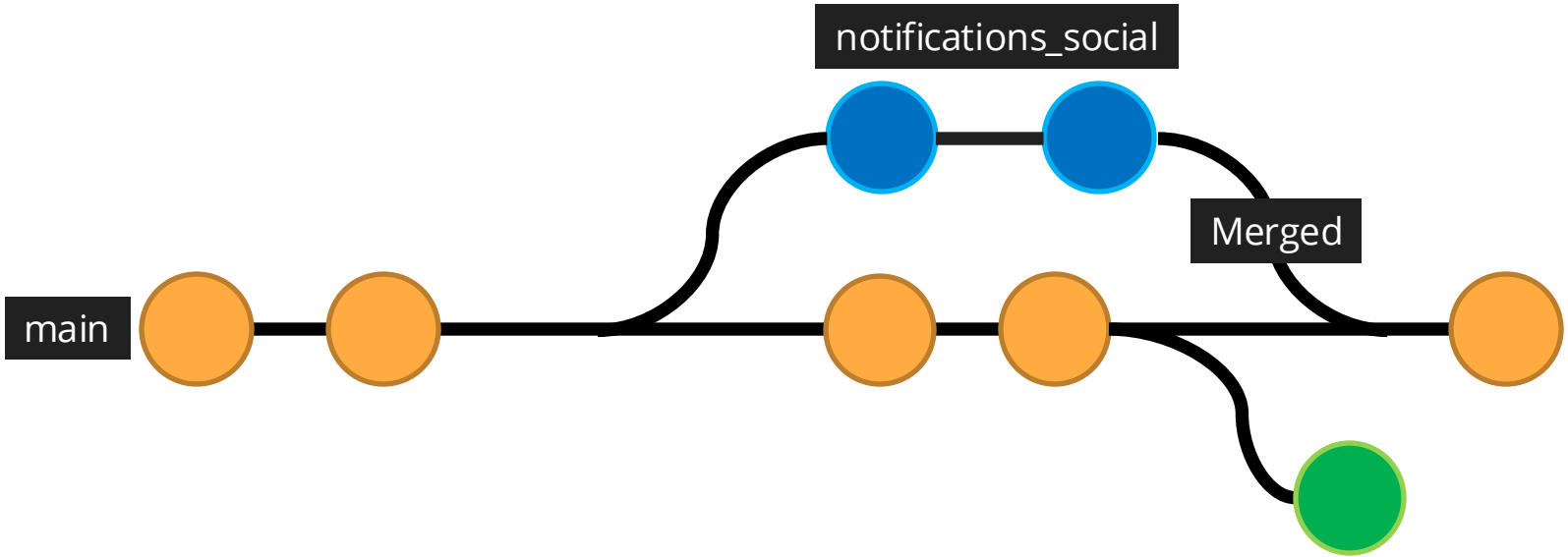
Force delete a branch even if there are unmerged changes.

```
$ git branch -D notifications_feature
```

Rename the currently active branch.

```
$ git branch -m notifications_social
```

Merging Branches



Use **git merge** to merge a branch into the **main** branch. If you're on the branch, checkout to main/master first and then merge.

```
$ git checkout main
$ git merge notifications_social
```

Activity

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1. Create a new directory for your project
2. Initialize a new Git repository
3. Create a new file and add some content
4. Add the file to the staging area
5. Commit the changes
6. Create a new branch
7. Switch to the new branch
8. Create a new file and add some content
9. Add the file to the staging area
10. Commit the changes
11. Merge the changes from the feature branch
12. View the commit history

Poll Time

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Q. Which Git command is used to view the commit history of a repository?

a. git log

b. git diff

c. git blame

d. git status



Poll Time

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Q. Which Git command is used to view the commit history of a repository?

a. **git log**

b. git diff

c. git blame

d. git status



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Activity

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1. Navigate to the repo you created in first part.
2. Create a branch called '**feature/payments-<your_name>-DCP121**' and switch to it.
3. Add a new file now called '**payments.js**' with the following content -> **console.log('added payments')**
4. Add another file called '**config.js**' with the following content -> **console.log('configs for the project resides here')**
5. Add both files to staging and then commit them.
6. Now switch to main branch and merge the newly created branch into main.
7. Delete the feature branch now.
8. Create another branch called '**hotfix/payments-<your_name>-PLT157**' and switch to it.
9. Go to '**payments.js**' file and add the content -> **console.log('fixed the bug')**
10. Repeat steps -> 5 to 7

Summary

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- git diff is used to compare two commits or branches.
- git revert is used to roll back changes. It creates another commit.
- Branching allows several team members to work on one codebase as they can develop features simultaneously. It also helps you to introduce checkpoints in the code.
- git checkout -b <branch_name> can be used to create and checkout to the new branch in one go.
- git branch –d allows you to delete a branch once it has been merged.
- git merge allows you to merge two branches.

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Session Feedback



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Next Session:
Collaborative Asset Management Using GitHub

Thank You

Please remember to complete
your assessments and learning for
the week on
PRISM

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